

# MODELING OF MICRO-DOPPLER SIGNATURES FROM ROTOR BLADES

Baoshuai Wang<sup>1</sup>, Weilu Li<sup>1</sup>, Lan Du<sup>1</sup>, Huiruo Shi<sup>1</sup>

<sup>1</sup>National Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China  
wbsatr@163.com, dulan@mail.xidian.edu.cn

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## Abstract

In this paper, we propose to establish the mathematical model for the rotor blades in a prop aircraft or a turbojet, where the targets' flying directions are vertical to the rotor blades' rotating plane. Experimental results with electromagnetic computation data obtained via a CADFEKO model prove the correctness of the proposed mathematical model. Based on the established mathematical model, we investigate the relationship between the micro-Doppler signatures and the azimuth angle. We wish our research reported here will provide researchers additional information which can be utilized as a complement to some existing target recognition methods.

## 1 Introduction

It is well known that moving targets will cause a Doppler frequency shift to the radar returned signal [1]. Recent years, V. C. Chen points out rotating or oscillating components on the targets tend to induce additional frequency modulations around the centre frequency of the target, which is referred to as micro-Doppler effect [2]. Micro-Doppler effect is widespread in the real world, e.g. the swing arms and legs of a walking human, the precession of space cone-shaped targets. Different kinds of targets will induce different types of micro-Doppler modulations in radar returns which can be regarded as the unique features, therefore researches on micro-Doppler feature extraction are widely studied in radar target recognition areas [3-6].

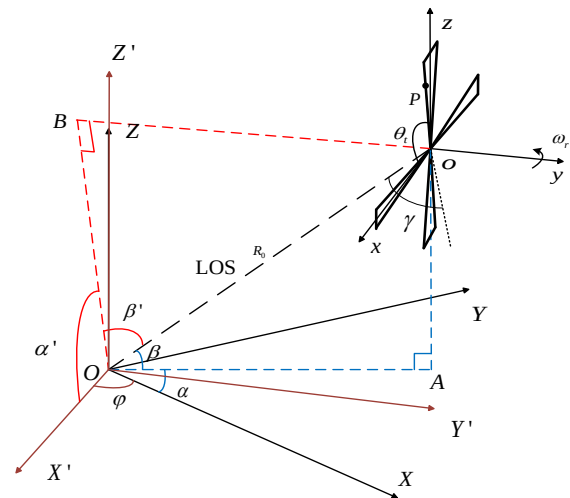
The jet engine modulation (JEM) induced by the high-speed rotating structures (rotor blades, propeller blades, or jet engine compressor and turbine blades) in an aircraft is a kind of micro-Doppler effect [7]. Most radar returns from airborne targets capture this modulation which is a form of frequency modulation and results in a number of uniformly located line spectrums about the centre frequency of the target. Many researches about the micro-Doppler signatures of helicopter are published recent years [8-11] while there are little literatures on prop aircraft and turbojet. There are mainly two reasons: 1) For a helicopter, its rotating plane is parallel to its flying direction which results in a simple mathematical model while for a prop aircraft or a turbojet the rotating planes are vertical to their flying directions and this will results in a more complex mathematical model; 2) Since a helicopter's

main rotor blade is usually much larger than the rotor blade of a prop aircraft or a turbojet, the Doppler modulation caused by the helicopter's main rotor is more easily to be captured in the radar returns.

However, experiment results with measured data from the three kinds of aircrafts (turbojet, prop aircraft and helicopter) show that the micro-Doppler modulation can also be captured in the radar returns for the prop aircraft and turbojet [6]. In some flying attitudes, the micro-Doppler modulations of a prop aircraft are even as visible as those of a helicopter [6]. Thus there is an exigent demand to establish the mathematical model for the prop aircraft or turbojet in order to help researchers obtain more detailed theoretical analysis. Since the flying directions are vertical to the rotating plane for both the prop aircraft and turbojet, in this paper we take the prop aircraft as an example to establish the mathematical model and the results are the same with the turbojet.

Based on the mathematical model for the main rotor blades in a helicopter, the research reported here proposes to establish the mathematical model for the prop aircraft. Then the correctness of the proposed mathematical model is verified using the electromagnetic computation data via a simple CADFEKO model. Finally, we investigate the relationship between the micro-Doppler signatures and the azimuth angle.

## 2 Mathematical Model of Rotating Structures in a Prop Aircraft



**Figure 1.** Geometrical relationship between radar and rotating structure when rotating plane is vertical to the flying direction

Figure 1 illustrates the geometry relationship between the radar and the rotating fan of which the rotating plane is vertical to the airplane's flying direction, i.e. the rotating plane (xoz) is vertical to the (XOY) plane of the radar coordinates (OXYZ). The rotor blades rotate about the y-axis in the (xoz) plane of the rotor's local coordinates (oxyz) with the rotation radian frequency denoting  $\omega_r$ .  $\alpha$  is the azimuth angle of the rotor to the radar line-of-sight (LOS) and  $\beta$  is the elevation of the rotor to the radar LOS. In order to utilize the approach in which the mathematical model for rotor blades in a helicopter is established, we formulate a reference coordinate (OX'Y'Z') which is parallel to the rotor's local coordinate (oxyz). The angle between the reference coordinate (OX'Y'Z') and the radar coordinate (OXYZ) is  $\varphi$ . Thus if we regard the (X'OY') plane as the reference plane and  $(\alpha', \beta')$  as the azimuth angle and pitching angle between the target and the radar LOS, i.e. the rotating plane (xoz) is parallel to the reference plane (X'OZ'), we can establish the mathematical model for rotor blades in a prop aircraft the same way as that in a helicopter [7].

Assume that there is a scatter P in a rotor blade rotating about the blade centre (o) with the rotation radian frequency  $\omega_r$ . The initial distance between the blade centre (o) with the radar is  $R_0$  and the initial distance between the blade centre and the scatter P is  $l$  at  $t = 0$ .

At  $t = t$ , the distance between the scatter and the radar is

$$s_B(t) = (L_2 - L_1) \sin c \left[ \frac{2\pi}{\lambda} (L_2 - L_1) \cos \beta' \cos(\theta_0 + \omega_r t - \alpha') \right] \cdot \exp \left[ -j \frac{4\pi}{\lambda} \left( R_0 + \frac{L_1 + L_2}{2} \cos \beta' \cos(\theta_0 + \omega_r t - \alpha') \right) \right] \cdot \exp [j 2\pi (f_0 - f_d) t] \quad (3)$$

For a rotor with  $N$  blades, the total received signal can be expressed as:

$$s_N(t) = \sum_{k=0}^{N-1} (L_2 - L_1) \sin c \left[ \frac{2\pi}{\lambda} (L_2 - L_1) \cos \beta' \cos(\theta_k + \omega_r t - \alpha') \right] \cdot \exp \left[ -j \frac{4\pi}{\lambda} \left( R_0 + \frac{L_1 + L_2}{2} \cos \beta' \cos(\theta_k + \omega_r t - \alpha') \right) \right] \cdot \exp [j 2\pi (f_0 - f_d) t] \quad (4)$$

where  $\theta_k = \theta_0 + 2\pi k / N$ ,  $k = 0, 1, \dots, N-1$  denote the initial rotation angle of the  $(k-1)$ th blade.

Now we have obtained the mathematical model for the rotor blades in a prop aircraft which is formulated in the reference coordinate (OX'Y'Z'). In order to obtain the corresponding mathematical model in the radar coordinate (OXYZ), we need to find the relationship between the two coordinate (OXYZ), and (OX'Y'Z'). For the rotation centre (o), its coordinate in the reference coordinate

$$R_p(t) = \sqrt{(R_t \cos \beta' \cos \alpha' + l \cos \theta_t)^2 + (R_t \sin \beta')^2 + (R_t \cos \beta' \sin \alpha' + l \sin \theta_t)^2} \approx R_0 + vt + l \cos \beta' \cos(\theta_0 + \omega_r t - \alpha') \quad (1)$$

where  $v$  denotes the radial velocity of the fuselage with respect to radar. Thus the radar received signal from P can be derived as:

$$s_p(t) = \exp \left[ -j \frac{4\pi}{\lambda} (R_0 + vt + l \cos \beta' \cos(\theta_0 + \omega_r t - \alpha')) \right] \cdot \exp(j 2\pi f_0 t) = \exp \left[ -j \frac{4\pi}{\lambda} (R_0 + l \cos \beta' \cos(\theta_0 + \omega_r t - \alpha')) \right] \cdot \exp[j 2\pi (f_0 - f_d) t] \quad (2)$$

where  $f_0$  and  $\lambda$  denote the carrier frequency and the wavelength of the radar transmitted signal respectively.

$f_d = \frac{2v}{\lambda}$  is the fuselage Doppler frequency.

In order to get the radar returns from the rotor blade P stays, we integrate (2) over the length of the blade  $L = L_2 - L_1$  with  $L_2$  and  $L_1$  denoting the distance between the blade tips and the centre of the rotation and the distance between the blade roots and the centre of the rotation, respectively. The radar returns from one rotor blade becomes:

is  $(R_t \cos \beta \cos(\alpha + \varphi), R_t \cos \beta \sin(\alpha + \varphi), R_t \sin \beta)$  while we can utilize  $\alpha'$  and  $\beta'$  to represent its coordinate as  $(R_t \cos \beta' \cos \alpha', R_t \sin \beta', R_t \cos \beta' \sin \alpha')$ . Then we can find the following equations:

$$\begin{cases} R_t \cos \beta \cos(\alpha + \varphi) = R_t \cos \beta' \cos \alpha' \\ R_t \cos \beta \sin(\alpha + \varphi) = R_t \sin \beta' \\ R_t \sin \beta = R_t \cos \beta' \sin \alpha' \end{cases} \quad (5)$$

We can obtain the relationship between  $(\alpha', \beta')$  and  $(\alpha, \beta, \varphi)$  as

$$\begin{cases} \alpha' = \arccot(\cos(\alpha + \varphi) \cot \beta) \\ \beta' = \gamma = \arcsin(\sin(\alpha + \varphi) \cos \beta) \end{cases} \quad (6)$$

Substituting equation (6) into equation (4) and compensating the motion and removing the constant phase term, we then obtain the final formation of the mathematical model for the rotor blades in a prop aircraft as:

$$s_N(t) = (L_2 - L_1) \sum_{k=0}^{N-1} \left[ \sin c \left[ \frac{2\pi}{\lambda} (L_2 - L_1) \cos(\arcsin(\sin(\alpha + \varphi) \cos \beta)) \right] \cos(\theta_k + \omega_r t - \arccot(\cos(\alpha + \varphi) \cot \beta)) \right] \cdot \exp \left[ -j \frac{4\pi}{\lambda} \left( R_0 + \frac{L_1 + L_2}{2} \cos(\arcsin(\sin(\alpha + \varphi) \cos \beta)) \right) \cos(\theta_k + \omega_r t - \arccot(\cos(\alpha + \varphi) \cot \beta)) \right] \quad (7)$$

We rewrite the mathematical model of a helicopter in equation(8) for a comparison with equation(7)

$$s_N(t) = \sum_{k=0}^{N-1} \left[ (L_2 - L_1) \sin c \left[ \frac{2\pi}{\lambda} (L_2 - L_1) \cos \beta \cos(\theta_k + \omega_r t - \alpha) \right] \exp \left[ -j \frac{4\pi}{\lambda} \left( R_0 + \frac{L_1 + L_2}{2} \cos \beta \cos(\theta_k + \omega_r t - \alpha) \right) \right] \right] \quad (8)$$

Comparing the above two equations, i.e. equation (7) and equation (8), we can see that if the pitching angle  $\beta$  is a constant, the differences between the two equations are caused by the azimuth angle  $\alpha$ . In equation (8),  $\alpha$  acts as an initial rotation angle while in equation (7),  $\alpha$  not only affects the initial rotation angle but also the coefficient that influences the width of  $\sin c(\bullet)$  function

### 3 Signal-Domain Signatures

In this part, we utilize the electromagnetic computation data obtained via the CADFEKO models shown in Figure 2 to evaluate the mathematical model established in Equation (8). Figure 2(a), (b) and (c) depict the simplified CADFEKO models for the rotating parts in a helicopter, prop aircraft and turbojet respectively. The rotation speeds and configuration parameters of the CADFEKO models shown in Figure 2(a), (b) and (c) are set according to the parameters referred to 'H-1', 'P-1' and 'T-1' in Table 1 respectively. Table 2 tabulates the scene parameters. We assume the radar works at S band with a carrier frequency of 2.85GHz. In order to evaluate the effects the azimuth angle has on the returned signal, in the simulation, we fix the elevation angle to  $30^\circ$ . The electromagnetic computation data are calculated via the physical optics (PO) method.

Using the CADFEKO models depicted in Figure 2 (a), (b) and (c), we can obtain the corresponding time-domain signal and Doppler spectra for the rotating structures in helicopter, prop aircraft and turbojet shown in Figure 3, Figure 4 and Figure 5, respectively. From Figure 3 we can see that the Doppler spectrum of helicopter does not varies much with the azimuth angle  $\alpha$ . This is due to the fact that in Equation (7)  $\alpha$  acts as an initial rotation angle which determines when the flash occurs in time domain while does not affect the Doppler spectra. From Figure 4 and Figure 5 we can see that the micro-Doppler signatures of prop aircraft and turbojet are affected much by the azimuth angle. As the azimuth changes from  $10^\circ$  to  $90^\circ$ , the width of the  $\sin c(\bullet)$  function gets wider while in the Doppler domain, the bandwidth of Doppler spectra, the num-

ber of spectra lines and the interval between the lines are also affected by the azimuth angle. This phenomenon is in accordance with our proposed mathematical model in Equation (7). Therefore, the results based on the electromagnetic computation data show that our proposed mathematical model is effective when dealing with the rotating structures in a prop aircraft or a turbojet where the rotating plane is vertical to its flying direction.

According to Equation (7), we obtain the simulated data using the MATLAB for the rotating structures in prop aircraft and turbojet depicted in Figure 7 and Figure 8. The rotation speeds, configuration parameters and typical scenes' parameters used here are the same as those used in Figure 4 and Figure 5 respectively. We can see that the simulated data shown in Figure 7 and Figure 8 are in accordance with the electromagnetic computation data.

### 4 Feature-Domain Signatures

In the real application, if the air defense radar system detects a target, the corresponding azimuth angle and the pitching angle between the target and the radar LOS can also be obtained at the same time. However, this useful information have not been used in the existing classification method for the three kinds of aircrafts [6,12]. According to the theoretical analysis and corresponding simulation results in part 2 and part 3, we note that for the same prop aircraft or turbojet, the Doppler spectrum varies with the azimuth angle. In this chapter, we will investigate the relationships between the commonly used micro-Doppler features and the azimuth angle. Three commonly used features are described as follows:

**Feature1:** the number of large eigenvalues that contain 98% of the eigenvalue spectrum's energy[12]:

$$\text{Feature1} = \arg \left( \sum_{i=1}^i \lambda_n^2 / \sum_{n=1}^T \lambda_n^2 = 0.98 \right)$$

where  $\lambda_n$  denotes the  $n$ th eigenvalue of the signal covariance matrix and  $T$  is the length of signal.

**Feature2:** the modified periodic factor[12]

$$\text{Feature2} = \zeta \cdot \nu$$

where  $\nu = 1 - T \cdot \sigma / (1 - T)$  denotes the normalized periodic factor,

$$\sigma = \lambda_1^2 / \sum_{n=1}^T \lambda_n^2$$

$\varepsilon_n = E_n / \sum_{n=1}^T E_n$ ,  $\zeta = -\sum_{n=1}^T \varepsilon_n \ln \varepsilon_n$  and  $\zeta = \zeta / \ln T$  represents the normalized energy entropy, where  $E_n$  denotes the  $n$ th sample's energy.

**Feature 3:** the entropy of the Doppler spectrum[6]

$$\text{Feature3} = -\sum_{f=1}^T p_f \ln p_f$$

where  $p_f = \frac{X(f)}{\sum_{f=1}^N X(f)}$ ,  $X = \text{FFT}(\mathbf{x})$  denotes the Doppler

spectra of the radar returns from the rotating blades.

Figure 9 shows the curves of the value of the above three features varying with the azimuth angle.

For each category of target, we utilize the parameters in Table 1 to simulate three types of aircraft and the corresponding scene parameters are tabulated in Table 2. The carrier frequency is 2.85GHz and dwell time is 150ms. From Figure 9, we can see that for the prop aircraft and turbojet the above three dimension feature vectors vary much with the azimuth angle due to the fact that the time domain signal and the corresponding Doppler spectra change with the azimuth angle. However the three dimension feature vector of helicopter does not change with the azimuth angle which is in accordance with the above

analysis in part 2 and part 3 that the azimuth angle only acts as an initial angle in the radar returns from the rotor blades in a helicopter and it does not change the time domain signal and the corresponding Doppler spectra.

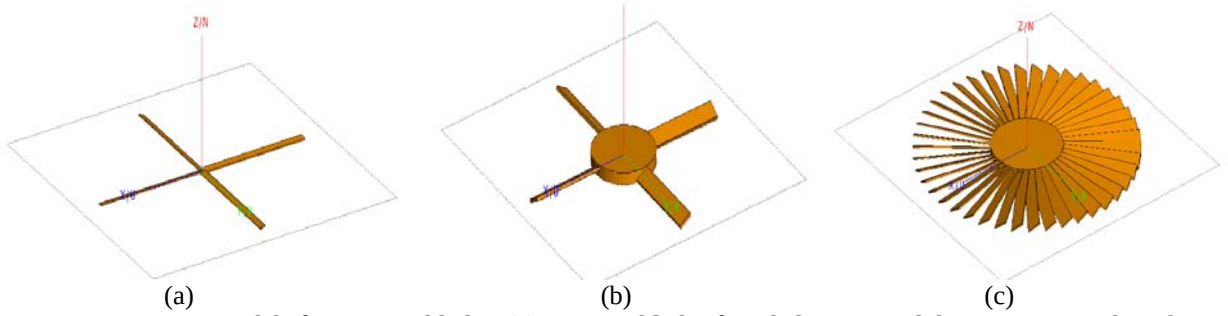
In addition, for the three dimension features of turbojet and prop aircraft, the azimuth angles where the value of each feature changes much are almost the same. Thus in the real application, we can divide the azimuth angles into different angle domains according to the inflexion where the value of features changes much. In each such azimuth angle domain, the lower dimension feature vector and simpler classifier can be utilized to realize the classification of the three kinds of aircraft. This is meaningful in the real application where real time classification lies at the heart of the classification scheme.

**Table 1:** Rotation speeds and configuration parameters of the rotating structures of 3 helicopters (H), 3 prop aircrafts (P) and 3 turbojets (T)

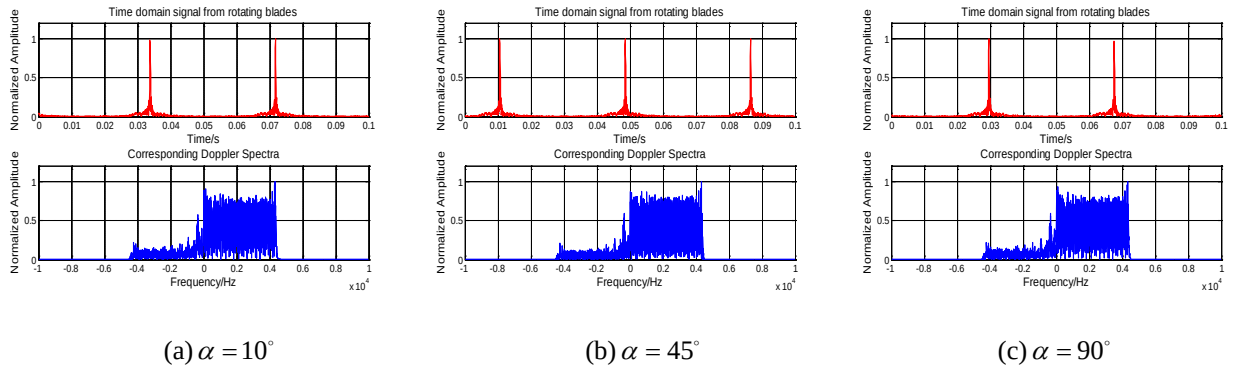
| Aircraft Index | Rotate speed $f_r$ (r/min) | $L_1$ (m) | $L_2$ (m) | Blade number $N$ | Aircraft Index | Rotate speed $f_r$ (r/min) | $L_1$ (m) | $L_2$ (m) | Blade number $N$ |
|----------------|----------------------------|-----------|-----------|------------------|----------------|----------------------------|-----------|-----------|------------------|
| H-1            | 395                        | 0         | 5.64      | 4                | P-3            | 1650                       | 0.12      | 1.15      | 5                |
| H-2            | 265                        | 0         | 7.8       | 4                | T-1            | 3000                       | 0.24      | 0.75      | 42               |
| H-3            | 185                        | 0         | 10.64     | 5                | T-2            | 3520                       | 0.38      | 1.1       | 38               |
| P-1            | 1900                       | 0.36      | 1.2       | 4                | T-3            | 8615                       | 0.18      | 0.51      | 27               |
| P-2            | 950                        | 0.28      | 1.91      | 6                |                |                            |           |           |                  |

**Table 2:** Typical scenes' parameters

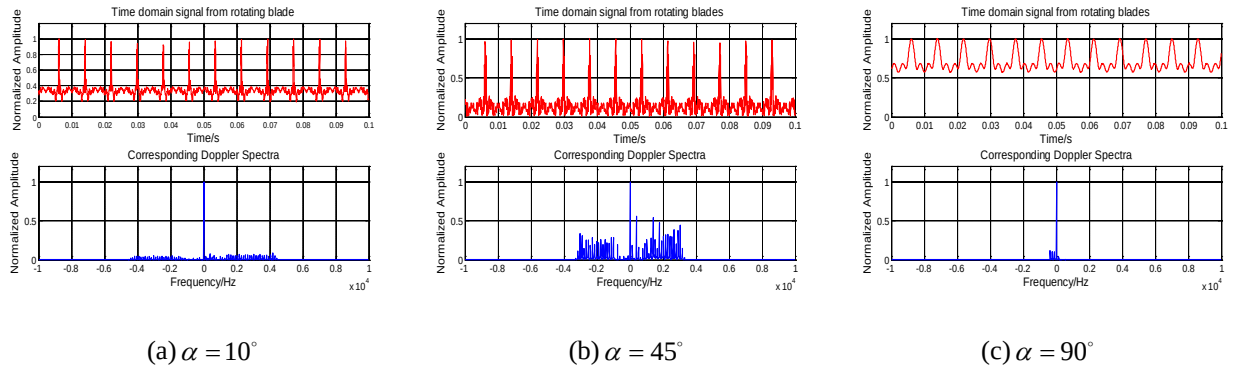
| Aircraft category | $\beta / ^\circ$ | $\theta_0 / ^\circ$ | $\varphi / ^\circ$ | $R_0 / \text{m}$ | $h / \text{m}$ | Placement manner |
|-------------------|------------------|---------------------|--------------------|------------------|----------------|------------------|
| Helicopter        | 30               | 20                  | 20                 | 10000            | 1000           | Horizontal       |
| Prop aircraft     | 30               | 20                  | 20                 | 10000            | 1000           | Vertical         |
| Turbojet          | 30               | 20                  | 20                 | 10000            | 1000           | Vertical         |



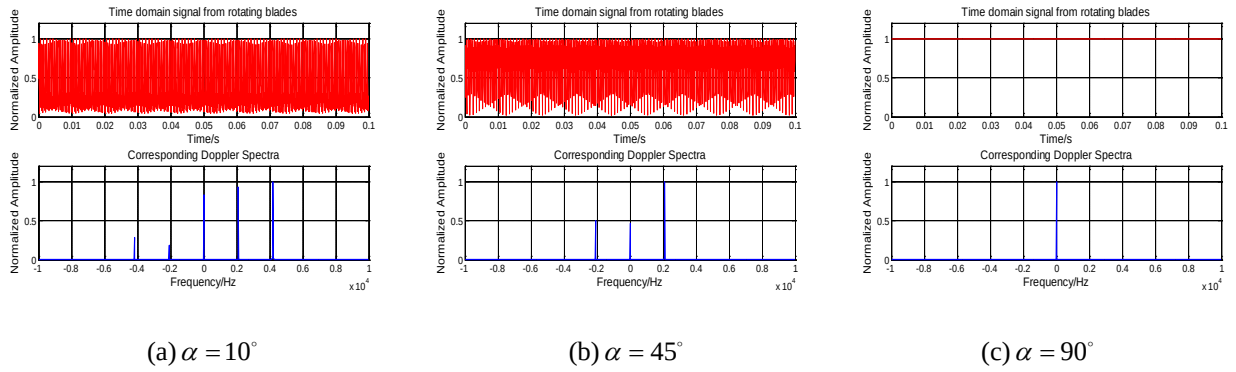
**Figure 2:** CADFEKO models for rotating blades. (a) rotating blades for a helicopter and the rotation speeds and configuration parameters are set according to the parameters referred to 'H-1' in Table 1; (b) rotating blades for a prop aircraft and the rotation speeds and configuration parameters are set according to the parameters referred to 'P-1' in Table 1; (c) rotating blades for a turbojet and the rotation speeds and configuration parameters are set according to the parameters referred to 'T-1' in Table 1.



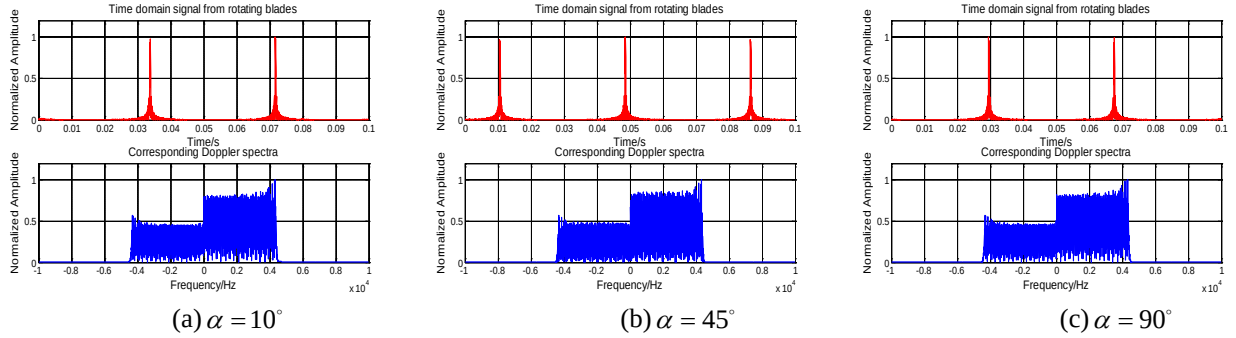
**Figure 3:** Radar return signal from the CADFEKO model depicted in **Figure 2(a)** under different azimuth angles.



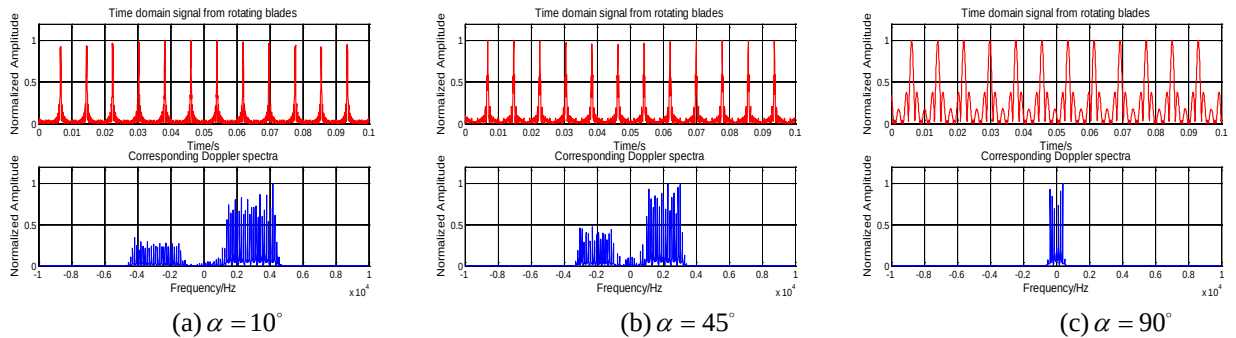
**Figure 4:** Radar return signal from the CADFEKO model depicted in **Figure 2(b)** under different azimuth angles.



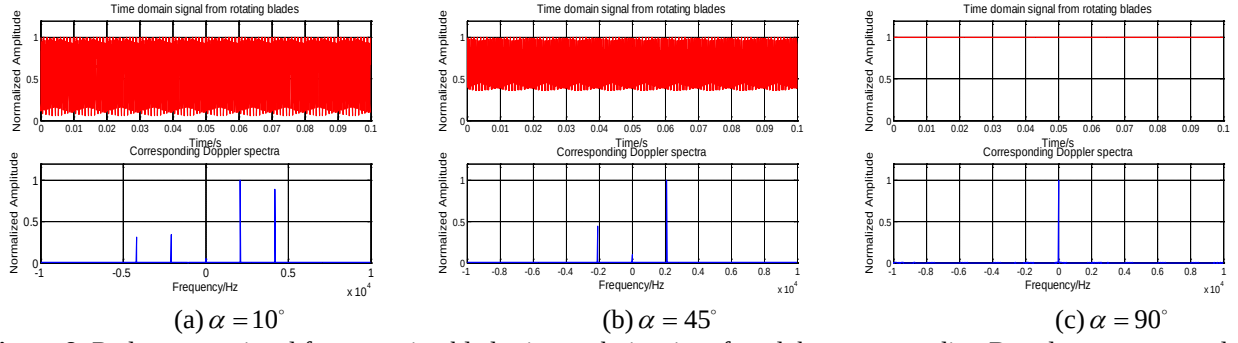
**Figure 5:** Radar return signal from the CADFEKO model depicted in **Figure 2(c)** under different azimuth angles.



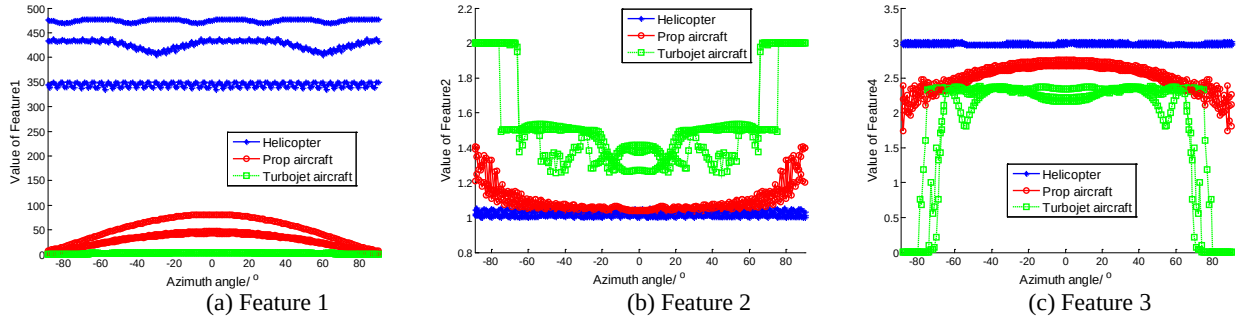
**Figure 6:** Radar return signal from rotating blades in a helicopter and the corresponding Doppler spectrum under different azimuth angles.



**Figure 7:** Radar return signal from rotating blades in a prop aircraft and the corresponding Doppler spectrum under different azimuth angles.



**Figure 8:** Radar return signal from rotating blades in a turbojet aircraft and the corresponding Doppler spectrum under different azimuth angles.



**Figure 9:** Variation of the values of different features with the azimuth angle.

## 5 Conclusion

In this paper, we establish the mathematical model for the prop aircraft and turbojet where the targets' flying directions are vertical to the rotor blades' rotation plane. The electromagnetic computation data obtained via a CADFEKO model prove that the proposed mathematical model is effective when dealing with the radar returns from the rotating structures in a prop aircraft or a turbojet. We investigate the relationship between the micro-Doppler signatures and the azimuth angle. This information can be utilized as a complement to some existing target recognition methods.

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